Inspection, Cleaning and Lubrication of 7x19 and 7x7 Control Cables – General Data

1. General

WARNING **THIS PROCEDURE USES MATERIALS THAT MAY BE DANGEROUS TO THE PERSONNEL'S HEALTH AND/OR THE ENVIRONMENT. SUFFICIENT PRECAUTIONS MUST BE TAKEN AT ANY STEP OF USE. IF NOT, THIS CAN CAUSE DAMAGE TO THE ENVIRONMENT AND DANGEROUS EFFECTS ON THE PERSONNEL'S HEALTH AND SAFETY.**

This standard practice specifies the procedures and requirements for wear and damage inspection of 7x19 and 7x7 control cables such as used for flight controls; cleaning and lubrication procedures are also provided. The control cables are of 7x19 construction type (7 strands, each strand with 19 wires) or 7x7 (7 strands, each strand with 7 wires). Control cable wear or breakage usually occurs where the cable passes through fairleads, over pulleys and quadrants. At manufacture, the wire rope is treated with an internal lubricating compound. This lubrication compound decreases internal friction, helps to prevent corrosion and adds to the life of the cable. For carbon steel non-jacketed cables, added lubrication and protection from corrosion is applied externally when the cable assembly is made and or installed.

A. Materials and Equipment

1) Materials

- IPA or Acetone
- Lubricant – MIL–PRF–81322 (Aeroshell 22)
- DS108, Diestone DLS, or Diestone HFP

2) Equipment

- Cloths – lint free
- Cloth – course weave
- 10 power magnifying glass
- Fibre brush

B. Safety Precautions

1) Use a cloth of sufficient thickness for protection of your hands when inspecting control cables for broken wires. Broken wires can cause injuries.

2) Sufficient ventilation must be provided when using solvent in confined areas. When applying solvent, operator will wear protective respiratory equipment.

3) Protective gloves will be worn to avoid skin contact with solvent. If contact occurs, wash the area thoroughly with soap and water.

4) Refer to the applicable manufacturer's Material Safety Data Sheet (MSDS) for specific safety data on any of the materials specified herein.

2. Cleaning

CAUTION **DO NOT USE VAPOR DEGREASING, STEAM CLEANING, OR SOLVENTS TO REMOVE CORROSION PREVENTIVE COMPOUNDS FROM THE OUTER SURFACES OF THE CONTROL CABLES. THESE PROCEDURES WILL ALSO REMOVE THE INTERNAL LUBRICANT.**

A. Clean the cables as follows:

1) Use a clean, dry, lint-free cloth to remove all dirt, debris, and old lubricant from the surface of the control cable.

2) When cables remain installed in the aircraft, all accessible lengths of the cable must be cleaned so that all cable strands over the full length of travel at fairleads, pulleys, quadrants, and cable drums can be inspected. Controls must be moved through their full range to allow cleaning of the cable where it contacts fairleads, pulleys, quadrants, and cable drums.

3) When cables are removed from the aircraft, clean the full length of the cable.

4) If corrosion preventive compound is present locally on surface of cables, pulleys, quadrants or cable drums, use a small amount of DS108, Diestone DLS, or Diestone HFP on a clean, lint-free cloth to remove compound.

Note Corrosion preventive compounds can be opaque and may prevent thorough inspection of cable.

3. Inspection/Check

CAUTION IF REMOVING OR INSTALLING CONTROL CABLES, DISCONNECT MAIN BATTERY (REFER TO 24-30-31, MAIN BATTERY SYSTEM – MAINTENANCE PRACTICES, PARA 2.), AVIONICS START BATTERY (IF INSTALLED, REFER TO 24-30-32, AVIONICS START BATTERY (MOD 6/2315) – GENERAL DATA, PARA 4.) AND EMERGENCY BATTERY (IF INSTALLED, REFER TO 33-50-01, EMERGENCY LIGHTING BATTERY – GENERAL DATA, PARA 1.) AND DISCONNECT GROUND POWER.

A. Cable Damage

1) No bruising, kinking, untwisting, bends or distortion on any control cable is permitted. Install new control cable when such defects are found.

B. External Wear

Refer to Figure 1 for details.

1) It is not practical to measure individual wires to determine wear; however, minor wear of individual wire appearing on the surface of any control cable is permissible.

Note Whenever cable wear is found, the cause for wear must be determined and corrected.

2) If wear on individual wire exceeds 40% of the original wire size, remove and discard cables and install new cables.

Note An indication of wear beyond 40% is the apparent blending of worn areas on the adjacent wires.

C. Broken Wires

Refer to Figure 2 for details.

1) Do an inspection for broken wires as follows:

Note Broken wires do not always extend from the lay of the cable. When broken wires or corrosion are suspected, remove the cable and carefully bend and twist the cable with your hands to make sure that all broken wires are found.

a) Rub a cloth along the length of the cable in both directions. Broken wires catch the cloth.

b) Use special care in areas where sections of the cable go around pulleys, quadrants, drums and through fairleads or pressure seals. Move controls through their full range to expose all cable surfaces adjacent to fairleads, pulleys, quadrants, and/or cable drums.

c) Pay close attention to the sections of the cable that are worn or have corrosion.

2) When broken wires are found, relieve cable tension and carefully examine the cable with a 10 power magnifying glass to make the full level of the damage known.

3) Replace a control cable when one or more of the conditions in Table 1 — Cable Broken Wire Limits are exceeded.

Note Flight control cables with broken wire(s) occurring in a wear area shall be considered unserviceable.

Table 1 Cable Broken Wire Limits

Cable Type	Location	Allowable Broken Wire
7x7 other than Flight Control Cable	Any	3 per one inch (25.4 mm) cable span
7x19 other than Flight Control Cable	Any	6 per one inch (25.4 mm) cable span
7x7 Flight Control Cable	Any	3 per one inch (25.4 mm) cable span, not to occur in consecutive inch spans
7x19 Flight Control Cable	At critical point	3 within the length of cable travel at pulleys, quadrants, drums, fairleads, fuselage cut-outs and within 12 inches (304.8 mm) of a cable end or terminal
	Away from critical point	6 per one inch (25.4 mm) cable span, not to occur in consecutive one inch (25.4 mm) spans.

D. Corrosion

CAUTION DO NOT USE METALLIC WOOL, WIRE BRUSHES, OR ABRASIVE MATERIALS TO REMOVE SURFACE CORROSION FROM CABLES. THESE MATERIALS WILL CAUSE CONTAMINATION THAT WILL RESULT IN SUBSEQUENT INTERNAL WEAR AND CORROSION.

1) Carefully examine the cable external surfaces for corrosion. Use special care when the conditions that follow occur:

a) There are cable runs through or near areas such as battery and baggage compartments, lavatories, and wheel wells.

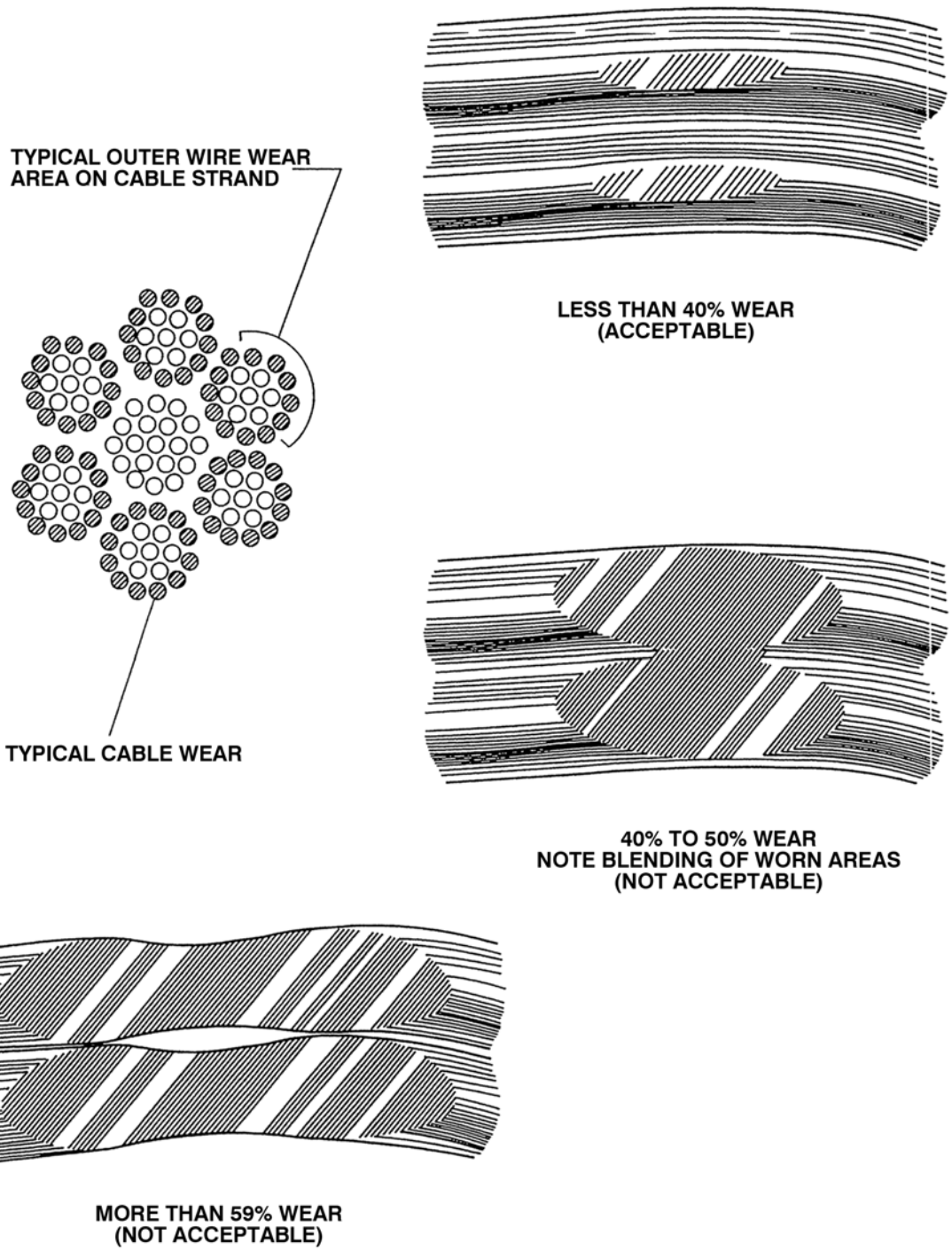
b) Broken wires are found in sections where wear is not caused by airframe components such as pulleys, pressure seals or fairleads.

c) Cleaning or metal brightening procedures are done adjacent to cable runs.

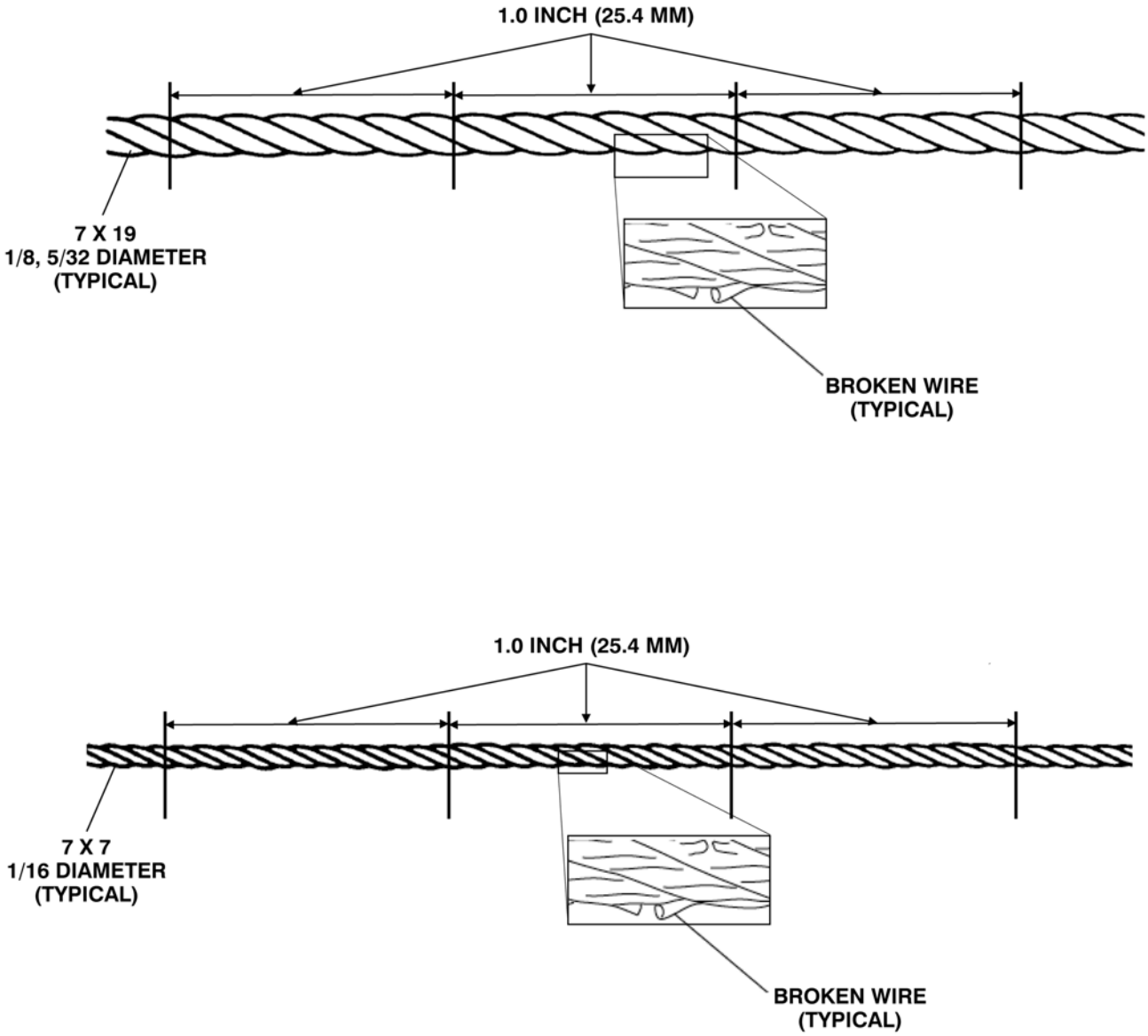
2) If external surface corrosion is found (white powder or a dark red), release the tension or remove the cable and twist it to visually examine the internal surfaces for corrosion products (white powder or a dark red).

3) Remove corrosion from the external surfaces with a coarse weave cloth or a fibre brush. After removing external corrosion and full cleaning to remove all debris, and corrosion particles, apply lubricant as per Para 4.

4) Replace control cable when internal corrosion is found.



Control Cable Wear Limits – Individual Wires
Figure 1



Control Cable Broken Wire Limits
Figure 2

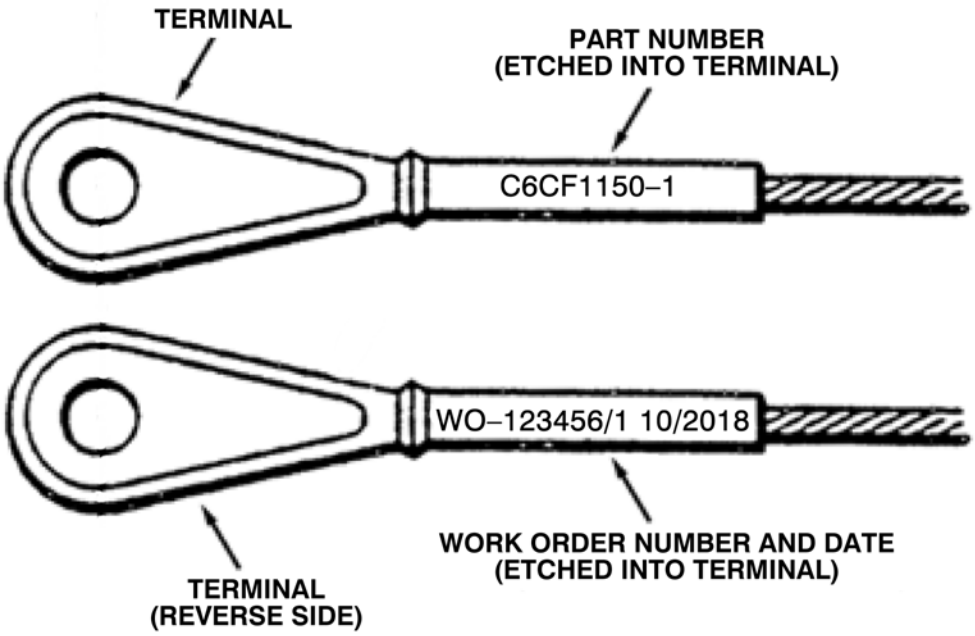
4. Lubrication

A. Examine the full length of carbon steel non-jacketed cables for a sufficient external coating of MIL–PRF–81322 (Aeroshell 22) lubricant. Use special care in those areas of the aircraft where frequent maintenance and servicing tasks are done.

Note This does not apply to corrosion resistant steel cables (composition B) and jacketed cables (Type 2). Lubrication attracts abrasive particles to corrosion resistant steel cables resulting in wear on pulleys, fairleads and pressure seals. Jacketed cables do not require lubrication. Cable composition can be confirmed by comparing the cable part number found on the cable end or terminal to the information in PSM 1-64-4. Refer to Figure 3.

B. When lubricant is removed as a result of inspection or removal and installation procedures on carbon steel non-jacketed cables, apply a thin layer of MIL–PRF–81322 (Aeroshell 22).

Note Apply lubricant MIL–PRF–81322 (Aeroshell 22) at regular times during the life of the carbon steel non-jacketed cable.



Control Cable Identification (Typical)
Figure 3

5. Requirements

Refer to Figure 4 for details.

A. Examine the airframe components.

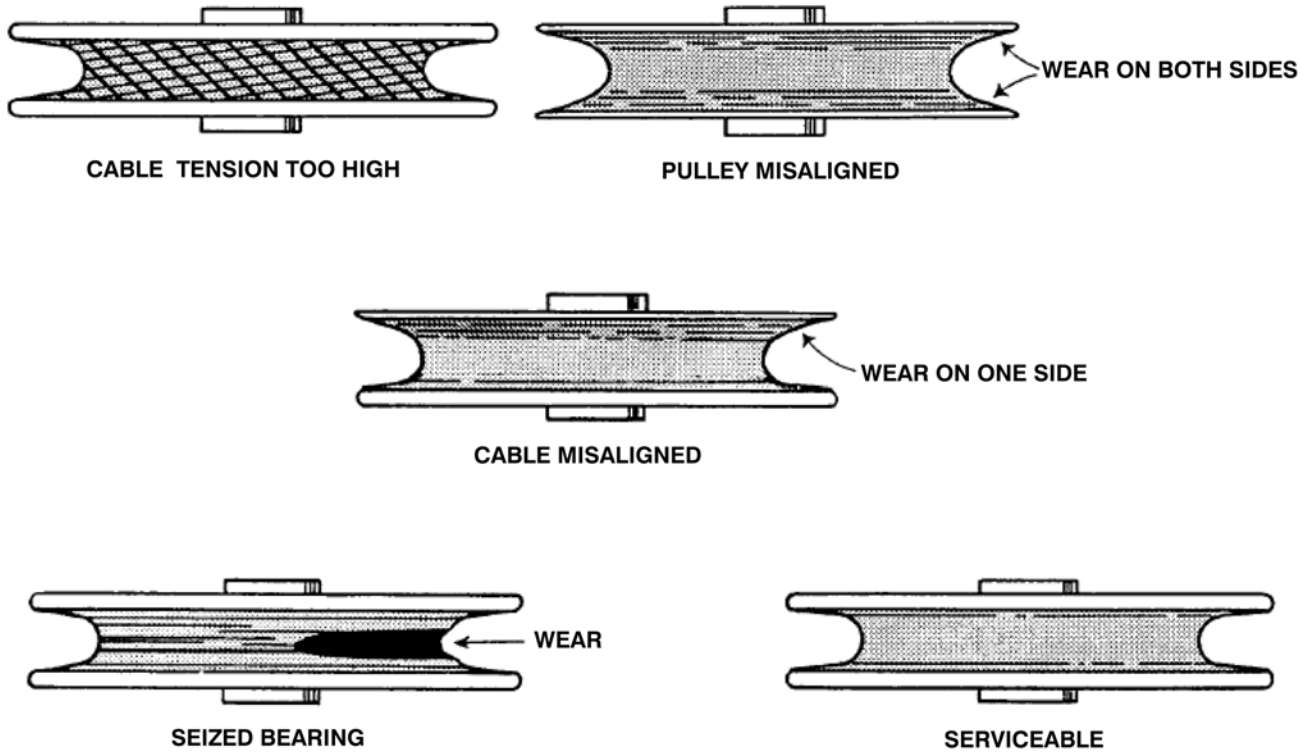
1) Examine the airframe components (such as pulleys, fairleads and guard pins) that are related to cables that have damage. Wear patterns on related airframe components are frequently an indication of the conditions that follow:

a) The cable tension is too high.

b) The pulley bearing is seized.

c) The airframe components not aligned correctly.

2) Correct all of the unsatisfactory conditions that can cause cable damage.



Pulley Wear Patterns (Typical)
Figure 4

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